

# CTR-GCN Two-Stream Baseline Reproduction: 3 Repeated Runs

NTU120 Interaction-26 (CSub) · joint and bone streams · one training per random seed, 3 seeds, 6 complete trainings · including per-class and confusion analysis · 2026-09-20

Two-stream fusion test Top-1, 3 repeated runs

91.20 ± 0.17

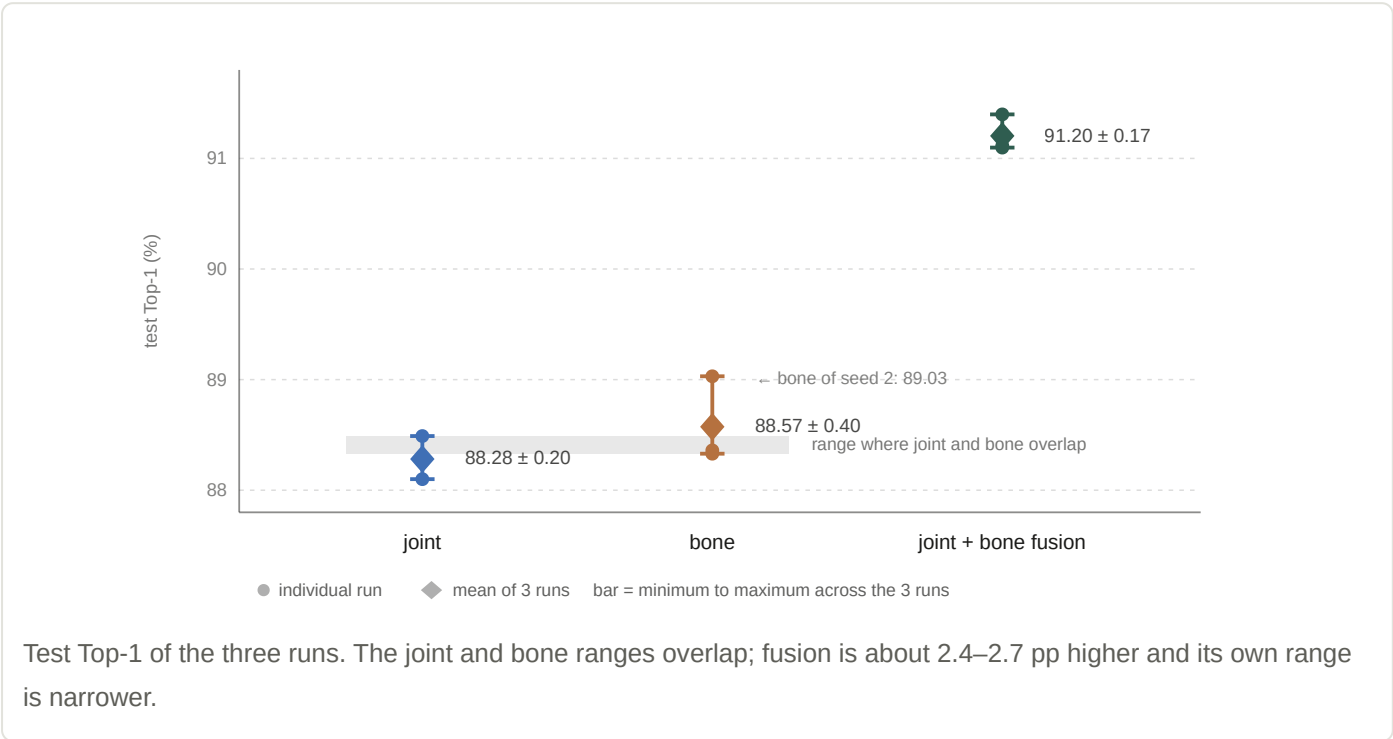
percentage points, range 0.30 pp

## 1 Experimental setup and fusion results

Under an identical configuration, the joint stream and the bone stream were each trained once per random seed for 3 seeds, giving 6 trainings in total, and fused pairwise at the score level, in order to measure the range this baseline takes across repeated runs. Fusion is the 1:1 equally weighted sum of the joint and bone scores.

Table 1 — Paired fusion result of each run (joint\_seedN + bone\_seedN)

| Seed      | Fusion test Top-1 | Fusion test Top-5 |
|-----------|-------------------|-------------------|
| seed 1    | 91.11%            | 98.82%            |
| seed 2    | 91.40%            | 98.87%            |
| seed 3    | 91.10%            | 98.84%            |
| Mean ± SD | 91.20 ± 0.17      | 98.84 ± 0.03      |



## 2 Single streams: joint and bone

joint is  $88.28 \pm 0.20$  pp (88.10–88.49) and bone is  $88.57 \pm 0.40$  pp (88.33–89.03); the two ranges overlap. The ordering of the two streams is not consistent across the three runs: in one run joint is 0.16 pp higher, in the other two bone is higher by 0.77 pp and 0.27 pp respectively. The 0.29 pp advantage in the bone mean comes from a single run (seed 2) — with that run removed, joint is 88.29 and bone is 88.34, a difference of 0.05 pp. The single-stream accuracies of the two streams are indistinguishable on this subset, and the ordering within any one run is not sufficient to decide between them.

Table 2 — Validation and test results of the 6 trainings (test Top-1 / Top-5 recomputed from the saved score files, digit-for-digit identical to the training logs)

| Seed                | Stream | Best val Top-1   | Best epoch | test Top-1       | test Top-5       |
|---------------------|--------|------------------|------------|------------------|------------------|
| seed 1              | joint  | 94.42%           | 60         | 88.49%           | 98.37%           |
| seed 1              | bone   | 93.81%           | 55         | 88.33%           | 98.20%           |
| seed 2              | joint  | 94.27%           | 56         | 88.26%           | 98.34%           |
| seed 2              | bone   | 94.42%           | 59         | 89.03%           | 98.28%           |
| seed 3              | joint  | 94.04%           | 54         | 88.10%           | 98.16%           |
| seed 3              | bone   | 93.58%           | 56         | 88.36%           | 98.38%           |
| joint mean $\pm$ SD |        | 94.24 $\pm$ 0.19 | 54–60      | 88.28 $\pm$ 0.20 | 98.29 $\pm$ 0.11 |
| bone mean $\pm$ SD  |        | 93.93 $\pm$ 0.43 | 55–59      | 88.57 $\pm$ 0.40 | 98.29 $\pm$ 0.09 |

## 3 Relation to the V1 report

The result of the seed 1 training was previously published as the V1 report. The median of the three fusion results (91.10 / 91.11 / 91.40) is 91.11, the same figure as in the V1 report, so that run sits in the middle of the three. The mean of 91.20 includes 91.11; the two do not constitute an improvement, they are the same baseline now carrying an error range.

Following section 2, the comparison in the V1 report stating that joint outperforms bone no longer holds; the main fusion figure and the evaluation method are unaffected.

## 4 Variance comparison

The standard deviation of fusion is 0.17 pp, below the 0.40 pp of the bone stream and the 0.20 pp of joint. The run that was high on a single stream (bone of seed 2, 0.46 pp above the bone mean) ends up only 0.29 pp above the fusion median after fusion. Relative to the better single stream of the same run, the fusion gains across the three runs are +2.62, +2.37 and +2.74 pp.

## 5 Validation set and test set

Validation accuracy is systematically higher than test accuracy: joint by 5.93–6.01 pp and bone by 5.22–5.48 pp, so the absolute values of the two are not directly comparable. In terms of ordering, the val ordering of the three joint runs ( $94.42 > 94.27 > 94.04$ ) matches their test ordering ( $88.49 > 88.26 > 88.10$ ); bone has one inversion of 0.03 pp (seed 1 is above seed 3 on val, while on test they are 88.33 against 88.36). The absolute val value is not a performance estimate, but its ordering signal is consistent with selecting the checkpoint by val Top-1.

## 6 Noise threshold

The three runs here differ only in the random seed, and the range of the fusion result is 0.30 pp. A fusion gain below 0.3 pp therefore cannot be distinguished from seed noise; deciding whether a change is effective requires comparing the means and ranges of two groups over multiple seeds, rather than comparing single results.

## 7 Per-class accuracy and per-class noise thresholds

The same criterion is far stricter at the class level. Each class has only 273–576 test samples, so the sampling standard deviation of a single-class accuracy,  $\sqrt{p(1-p)/n}$ , is 0.21–1.99 pp (median 1.13 pp), giving a  $2\sigma$  threshold of 0.42–3.97 pp that varies with the accuracy and the sample count of the class. The observed per-class ranges across the three runs are of the same magnitude: joint median 2.08 pp and maximum 7.83 pp, bone median 2.43 pp and maximum 6.52 pp, fusion median 1.56 pp and maximum 3.99 pp.

The overall 0.3 pp threshold therefore does not apply here. Every class-level comparison below is checked against that class's own  $2\sigma$  threshold; anything that does not exceed it is labelled noise and is not interpreted.

Table 3 — Per-class test accuracy (mean of the 3 runs), sorted by fusion mean in ascending order

| Class    | joint mean | bone mean | fusion mean | fusion range over 3 runs (pp) |
|----------|------------|-----------|-------------|-------------------------------|
| Class 11 | 59.25%     | 63.07%    | 65.28%      | 3.13                          |
| Class 12 | 61.69%     | 63.48%    | 67.19%      | 3.30                          |
| Class 15 | 64.12%     | 72.23%    | 75.83%      | 2.96                          |
| Class 6  | 86.23%     | 88.65%    | 89.37%      | 1.45                          |
| Class 22 | 89.04%     | 84.00%    | 90.32%      | 0.87                          |
| Class 4  | 86.72%     | 87.08%    | 91.18%      | 1.45                          |
| Class 14 | 87.19%     | 87.42%    | 91.54%      | 2.43                          |
| Class 0  | 88.44%     | 87.47%    | 91.61%      | 3.65                          |
| Class 13 | 87.79%     | 86.98%    | 91.96%      | 3.65                          |
| Class 1  | 92.51%     | 89.73%    | 93.24%      | 2.90                          |
| Class 3  | 89.73%     | 88.41%    | 93.24%      | 2.54                          |
| Class 24 | 91.54%     | 90.61%    | 93.51%      | 2.43                          |
| Class 23 | 92.06%     | 93.97%    | 94.26%      | 2.43                          |
| Class 21 | 91.96%     | 91.20%    | 94.68%      | 3.99                          |
| Class 16 | 92.81%     | 92.17%    | 95.01%      | 1.22                          |
| Class 7  | 92.48%     | 93.21%    | 95.15%      | 2.55                          |
| Class 10 | 95.53%     | 89.25%    | 95.17%      | 1.09                          |
| Class 20 | 93.23%     | 93.06%    | 95.72%      | 1.56                          |
| Class 2  | 96.01%     | 93.60%    | 96.50%      | 1.45                          |
| Class 8  | 95.65%     | 95.53%    | 96.50%      | 1.09                          |
| Class 25 | 96.93%     | 97.63%    | 97.63%      | 0.69                          |
| Class 19 | 97.11%     | 98.67%    | 98.44%      | 0.52                          |
| Class 17 | 96.76%     | 98.15%    | 98.90%      | 0.52                          |
| Class 5  | 99.39%     | 98.66%    | 99.51%      | 1.09                          |
| Class 18 | 98.32%     | 99.01%    | 99.59%      | 0.35                          |
| Class 9  | 99.51%     | 98.17%    | 99.88%      | 0.37                          |

Classes 11, 12 and 15 are the three hardest (fusion means 65.28%, 67.19% and 75.83%) and sit well apart from the other 23; the highest classes, 9, 18 and 5, are all above 99%.

## 8 Confusion structure

The figure below is the confusion matrix of fusion, normalised by row (each row divided by the sample count of that class, removing the differences in class size), averaged over the three runs, with the diagonal set to zero so that the destinations of misclassifications are visible.

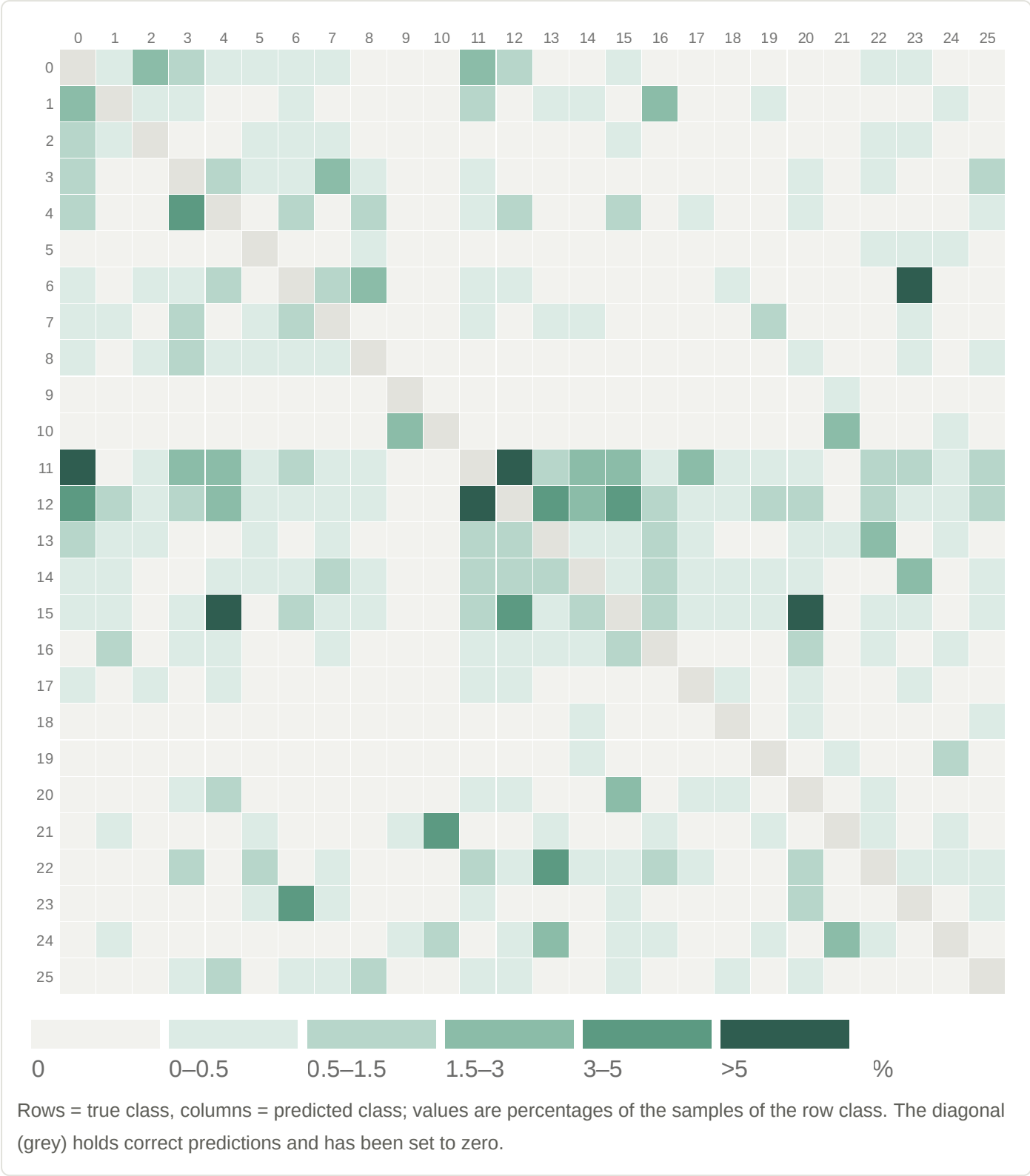
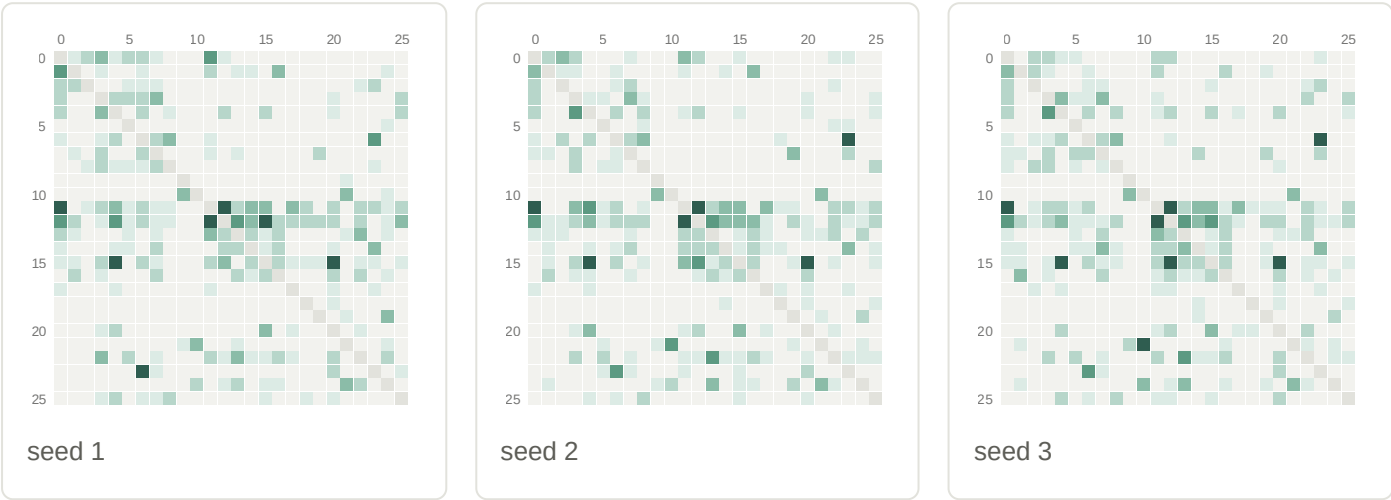


Table 4 — Most confused directions (fusion, row-normalised confusion averaged over 3 runs, top 15 off-diagonal)

| Direction (true → predicted) | Share of that class | Approx. samples | Bidirectional |
|------------------------------|---------------------|-----------------|---------------|
| Class 11 → Class 12          | 9.57%               | 55              | Yes           |
| Class 11 → Class 0           | 9.45%               | 54              | —             |
| Class 12 → Class 11          | 8.16%               | 47              | Yes           |
| Class 15 → Class 4           | 7.42%               | 43              | —             |
| Class 15 → Class 20          | 6.67%               | 38              | —             |
| Class 6 → Class 23           | 5.07%               | 14              | Yes           |
| Class 23 → Class 6           | 4.75%               | 27              | Yes           |
| Class 21 → Class 10          | 4.28%               | 25              | —             |
| Class 15 → Class 12          | 4.00%               | 23              | Yes           |
| Class 12 → Class 15          | 3.82%               | 22              | Yes           |
| Class 12 → Class 0           | 3.59%               | 21              | —             |
| Class 12 → Class 13          | 3.53%               | 20              | —             |
| Class 4 → Class 3            | 3.38%               | 9               | —             |
| Class 22 → Class 13          | 3.19%               | 18              | —             |
| Class 12 → Class 4           | 2.89%               | 17              | —             |

Misclassification is asymmetric, so the table lists each direction separately. Three pairs are bidirectional: classes 11 and 12, classes 12 and 15, and classes 6 and 23. The errors of class 11 flow both to class 12 (9.57%) and to class 0 (9.45%), making it the heaviest class in two directions at once.



The matrices of the individual runs (same colour scale) share the same structure, so the confusion pattern above does not change with the seed.

## 9 Effect of fusion at the class level

Class by class, fusion is compared against max(joint, bone) using the mean of the 3 runs and checked against the  $2\sigma$  threshold of each class.

Table 5 — Per-class change of fusion relative to max(joint, bone) (pp, mean of 3 runs); the 6 largest gains and the 6 largest drops

| Class    | Mean change | The 3 runs individually | $2\sigma$ threshold | Verdict                  |
|----------|-------------|-------------------------|---------------------|--------------------------|
| Class 13 | +4.17       | +3.99 / +4.17 / +3.12   | 2.27                | Above threshold          |
| Class 14 | +4.12       | +4.70 / +3.30 / +4.35   | 2.32                | Above threshold          |
| Class 4  | +4.11       | +3.26 / +3.26 / +4.35   | 3.41                | Above threshold          |
| Class 12 | +3.70       | +2.60 / +2.78 / +2.78   | 3.91                | Within threshold (noise) |
| Class 15 | +3.59       | +1.74 / +5.57 / +3.48   | 3.57                | Above threshold          |
| Class 3  | +3.50       | +1.45 / +3.62 / +4.71   | 3.02                | Above threshold          |
| Class 9  | +0.37       | +0.00 / +1.10 / +0.00   | 0.42                | Within threshold (noise) |
| Class 23 | +0.29       | +0.70 / -0.35 / +0.52   | 1.94                | Within threshold (noise) |
| Class 5  | +0.12       | +0.00 / -0.36 / +0.36   | 0.84                | Within threshold (noise) |
| Class 25 | +0.00       | +0.35 / -0.17 / -0.17   | 1.27                | Within threshold (noise) |
| Class 19 | -0.23       | -0.35 / +0.00 / -0.35   | 1.03                | Within threshold (noise) |
| Class 10 | -0.36       | -1.81 / +1.45 / -0.72   | 2.58                | Within threshold (noise) |

The gains that exceed the threshold are classes 13, 14, 4, 15 and 3 (+4.17, +4.12, +4.11, +3.59 and +3.50 pp), where class 15 only barely exceeds it (3.59 against a threshold of 3.57); the +3.70 pp of class 12 does not reach its threshold of 3.91 pp. On the side of drops, no class exceeds its threshold: the largest in absolute value is class 10 at -0.36 pp against a threshold of 2.58 pp, with the three runs giving -1.81, +1.45 and -0.72 pp, which do not even agree in sign. **Averaged over the three runs, there is no evidence that fusion makes any class worse.**

The three drops reported from the single V1 run do not survive averaging over three runs: class 10 was -1.81 pp in that single run against a three-run mean of -0.36 pp (threshold 2.58); class 2 was -1.45 pp against a mean of +0.48 pp (threshold 2.21); class 19 was -0.35 pp against a mean of -0.23 pp (threshold 1.03). All three are within the noise range, so these three conclusions in V1 do not hold.

The fusion gain correlates positively with the **disagreement rate** between the joint and bone streams (the share of samples on which the two streams predict differently, 1.8%–40.1% across classes, median 11.4%), with a Pearson correlation of  $r = 0.62$ , while it is essentially unrelated to the **difference** in the

accuracies of the two streams ( $r = -0.10$ ). That is, the benefit of fusion comes from the classes on which the two streams disagree, not from which stream is stronger on a class.

## 10 Limitations

- The variation measured here is a lower bound on the true variation. The DataLoader's `worker_init_fn` sets the random seed of each worker to the worker index 0–3, independently of `--seed`, so the randomness of the data augmentation (random rotation, random cropping) is identical across the three runs and only the model initialisation and the shuffling order change with the seed. This behaviour comes from the original CTR-GCN and was not introduced by this project.
- With only 3 repetitions the standard deviation is itself quite uncertain and can only be used to judge differences on the order of 0.3 pp, which is why the range is given alongside it.
- The fusion of all 9 joint-bone cross combinations was also computed, but picking the highest of the nine amounts to selecting a model on the test set, which violates the evaluation protocol, so it is not reported.
- The noise threshold of a per-class result is far higher than the overall one. The per-class  $2\sigma$  thresholds are 0.42–3.97 pp, so the overall 0.3 pp figure must not be used to judge per-class numbers.
- Classes keep their numeric indices 0–25. Neither the script that generated this 26-class subset nor its documentation is present on this server, so the action name behind an individual index cannot be confirmed; the per-class sample counts recomputed from `statistics/label.txt` together with the CSub protocol do confirm that the subset is the 26 classes A50–A60 and A106–A120, but counts repeat across classes and are not sufficient to pin down each index, so action names are not used.
- The conclusions apply only to the NTU120 Interaction-26 (CSub) subset and the current hyperparameters.

## 11 Method and verification

All 6 trainings used the same configuration (device [0,1], two GPUs, 65 epochs, base\_lr 0.1, step [35,55]); only `--seed` and `--work-dir` differed, and no yaml or code was modified. Each training took about 1 hour 55 minutes, and the total durations of the 6 runs differ by less than 1 minute.

The metrics of seed 1 were recomputed by the analysis script from its existing artifacts; the joint, bone and fusion figures are identical to those in the V1 report's `metrics_summary.json`; across the two directories of that run, the mtime and size of all 228 files and the md5 of the key files were unchanged before and after the new trainings.

Result files: `reports/seedvar_s2s3/seedvar_results_seeds1_2_3.json` (the single source of every number in this report), `seedvar_results_seeds1_2_3.md` and `analyze_seedvar.py`.

Training artifacts:

`work_dir/ntu120_interaction26/csub/ctr_gcn_{joint,bone}_server{,_seed2,_seed3}/`; the two directories without a suffix are seed 1 (the run published as V1) and were not modified.

```
cd /mnt/data/6008/CTR-GCN
python main.py --config config/ntu120-interaction26-csub/joint_server.yaml --phase train \
  --seed 2 --work-dir ./work_dir/ntu120_interaction26/csub/ctr_gcn_joint_server_seed2
python reports/seedvar_s2s3/analyze_seedvar.py 1 2 3
```

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6 complete trainings (3 seeds  $\times$  2 streams, 65 epochs each). All differences are expressed in percentage points (pp). With 3 repetitions, no significance test was performed.